

Improving Student Outcomes through Adaptive Learning Platforms: Experimental Evidence from the Dominican Republic

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Abstract

Most students in developing countries receive grade-level instruction despite lacking prerequisite skills, a mismatch that widens learning gaps as students progress. This paper reports evidence from a randomized controlled trial across 38 classrooms and 1,333 ninth-grade students in the Dominican Republic, in which classrooms were randomly assigned to one of three arms: computer-adaptive learning (CAL) software replacing two of seven weekly mathematics hours, CAL combined with small-group tutoring, or a business-as-usual control. CAL improved test scores by 0.29–0.31 standard deviations over six months, among the larger effects documented for education technology interventions in developing countries. Adding small-group tutoring yielded no detectable improvement over CAL alone; if anything, classrooms assigned to both interventions showed smaller gains than those receiving CAL alone, though this difference is not statistically significant. Evidence on mechanisms indicates that tutoring classrooms exhibited reduced engagement with the CAL platform. These results suggest that computer-adaptive learning software can be effectively integrated into classroom instruction to improve learning at scale, but that combining multiple interventions is not guaranteed to produce additive benefits and may instead generate unintended behavioral responses that offset their effectiveness.

JEL Classification: C93, I21, I28, J24, O15

Keywords: Computer-adaptive learning, education technology, randomized controlled trial, mathematics education, developing countries, Dominican Republic

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1 Introduction

Most students in developing countries receive grade-level instruction despite lacking prerequisite skills, and this mismatch widens learning gaps over time (Muralidharan et al., 2019). The problem is widespread: less than 20 percent of students in low- and middle-income countries meet minimum proficiency standards in mathematics (World Bank, 2017), and the typical student in Latin America performs approximately five years behind their OECD counterparts (IDB, 2023). When students who lack foundational knowledge are taught content beyond their current level, they fall further behind rather than catch up, contributing to both poor achievement and high dropout rates.

Computer-adaptive learning (CAL) platforms offer a potentially scalable solution to this instructional mismatch. By continuously assessing student knowledge and adjusting content difficulty accordingly, CAL can deliver personalized instruction at a fraction of the cost of one-on-one tutoring. Recent evidence from India demonstrates that CAL can generate substantial learning gains when substituted for regular classroom instruction, with effects ranging from 0.20–0.22 standard deviations at scale (?) to 0.45 standard deviations under intensive implementation support (Oreopoulos et al., 2026). Yet as policymakers seek to maximize learning recovery, a critical question remains unexplored: can combining CAL with other effective interventions—such as small-group tutoring—generate complementary benefits, or might such combinations produce unintended interference?

This paper addresses this question through a randomized controlled trial integrating CAL into regular mathematics instruction for 1,333 ninth-grade students across 38 classrooms in the Dominican Republic. We employed ALEKS, a CAL platform developed by McGraw Hill and adapted to the Dominican Republic’s math curriculum.^{footnote} The intervention was implemented by the Ministry of Education from 2018 to 2023, with support from the World Bank. All students in treated classrooms received licenses to access the software during two of their seven weekly math instruction hours, replacing rather than supplementing traditional instruction. To test for complementarities between technology-based and traditional instructional approaches, we implemented a second randomization within treated classrooms, providing up to three weekly hours of small-group tutoring to academically at-risk students. Tutoring sessions were scheduled during existing extracurricular periods to avoid confounding effects from additional instructional time.

We find that the use of the adaptive software during math classes improved student outcomes by 0.29 to 0.31 standard deviations by endline (six months after the start of the intervention). These positive effects are robust across model specifications and consistent across mathematical domains, including algebra and geometry, and across difficulty levels, including content below and at grade level. This places our estimate close to the upper end of effects documented for CAL interventions. Adding small-group tutoring did not enhance these gains; if anything, classrooms assigned to both interventions showed smaller effects than CAL alone, though the difference is not statistically significant.

Why did the two treatment arms have different impacts? Implementation fidelity does not explain the difference: both CAL and tutoring faced similar compliance rates (approximately 65–68 percent of intended dosage) and similar barriers, predominantly school closures and infrastructure failures. Instead, the evidence points to behavioral mechanisms. Students in tutoring classrooms—including non-tutees—were less likely to use the adaptive platform and exhibited lower levels of locus of control (i.e., belief in their ability to influence their own outcomes), compared to the CAL-only group. We also find evidence indicating that the tutoring may not have been well-targeted, as students selected by teachers to receive tutoring were not among the lowest performers at baseline, as was intended by the intervention.

This paper makes three contributions to the literature on educational technology in developing countries. First, we provide experimental evidence on CAL as a *substitute* for, rather than a supplement to, traditional instruction. Most existing studies have evaluated CAL in after-school settings or combined with tutoring (Muralidharan et al., 2019; Guryan et al., 2023), leaving open questions about its effectiveness when integrated into regular school schedules or implemented alone (Muralidharan et al., 2019). Recent evidence from India demonstrates that intensive implementation support can generate CAL effects of 0.45 standard deviations when dedicated personnel ensure high-fidelity platform usage (Oreopoulos et al., 2026), but such models require dedicated staff and face scalability constraints; the impact of CAL alone versus additional instructional time remains difficult to disentangle in prior designs (Glewwe and Muralidharan, 2016; Muralidharan and Singh, 2025). By replacing two of seven weekly math instruction hours with CAL sessions, we add to a small but growing body of evidence on its efficacy when implemented within the standard school curriculum, rather than as an after-school add-on (Muralidharan and Singh, 2025).

Second, we provide a rare experimental test of complementarities between technology-based and traditional instruction. Tutoring is among the most effective interventions for improving learning outcomes, with recent meta-analysis reporting an average effect size of 0.37 standard deviations Nickow et al. (2020), but its scalability is constrained by high costs and the difficulty of recruiting qualified tutors (Kraft and Falken, 2020; Nickow et al., 2020; Kraft et al., 2024). Small-group tutoring and CAL have each been proposed as more scalable alternatives to one-on-one models (Augustine et al., 2016; Guryan et al., 2023), and a natural policy response is to combine them, on the assumption that combining the two would yield benefits beyond either intervention alone. Yet this assumption has rarely been tested directly. Our two-stage randomization design isolates the CAL×Tutoring interaction, providing causal evidence on whether adaptive technology and human tutoring are complementary or competing inputs into student learning.

Third, we document behavioral mechanisms through which intervention complementarities may fail. Contrary to expectations, combining CAL with in-person tutoring did not reinforce its effects; instead, the combined treatment produced gains indistinguishable from zero relative to the control group. We show this pattern is associated with reduced platform engagement and lower locus of

control among students in tutoring classrooms, including those not directly tutored, suggesting that the introduction of human support may erode the self-directed engagement that drives CAL’s effectiveness. These findings underscore the importance of testing intervention bundles empirically, rather than assuming their learning effects are additive.

The deepening of the global learning crisis following COVID-19 highlights the need for more resilient, scalable approaches to instruction. Our findings show that adaptive learning technology can deliver substantial learning gains even amid the infrastructure constraints typical of developing-country settings. At the same time, the reduced effectiveness we observe when combining CAL with tutoring—linked to decreased platform engagement and shifts in student agency—shows that layering interventions does not guarantee additive effects, and that successful implementation requires understanding how components interact, not simply how many are deployed.

This paper proceeds as follows. In Section 2, we describe the context, intervention, and implementation. In Section 3, we describe our data. In Section 4 and 5, we present the empirical framework and results. Lastly, in Section 6, we discuss mechanisms, and in Section 7, we conclude.

2 Setting and Experimental Design

2.1 Background

The Dominican Republic exemplifies the mathematics learning crisis facing much of Latin America and the Caribbean. In 2022, the country ranked third to last globally on the OECD’s Programme for International Student Assessment (PISA), with 92 percent of students classified as low-performing in mathematics (OECD, 2023a,b). The consequences of these learning deficits become particularly acute during the transition to secondary education. The Dominican education system progresses from primary (grades 1-6) through lower secondary (grades 7-9) to upper secondary (grades 10-12), with students typically entering lower secondary around age 12.¹ Previous research indicates that up to 40 percent of students drop out between 8th and 12th grade (IDEICE, 2017).

Ninth grade represents a critical intervention point in this trajectory. Students at this stage have likely accumulated substantial learning deficits that threaten their ability to complete secondary education, yet they remain within the compulsory education system where remedial interventions can still be effectively implemented. Our study evaluates an intervention designed to address these challenges by providing targeted remedial mathematics training to ninth-grade students. To the extent that poor academic performance drives dropout decisions, improving learning outcomes may also promote educational retention; however, if students face binding non-academic constraints—such as economic pressures or family obligations—learning gains alone may prove insufficient to affect persistence. This study contributes to broader efforts to understand both pathways in the

¹The school year runs from late August to late June, organized over two academic semesters. The 2023-24 school year officially started on August 28th 2023 and ended on June 21st 2024.

Dominican Republic and the wider LAC region.

2.2 Intervention

The study and intervention was implemented in partnership with the Ministry of Education of the Dominican Republic (MINERD) and the Abdul Latif Jameel Poverty Action Lab (J-PAL).² The intervention was implemented between October 2023 and May of 2024 in a sample of 38 9th grade classrooms in 9 schools.³ The program had two core elements: a CAL intervention and a tutoring component. The first component introduced the use of a CAL platform (ALEKS) for two hours per week as part of schools' regular schedule, substituting two out of seven weekly hours scheduled for math instruction. Half of the classrooms were randomly selected to receive CAL licenses for all students and their math teachers (stratified by classroom size). The second component introduced small-group tutoring during extracurricular hours for approximately 20 percent of students in randomly selected classrooms from the first treatment group. Each component is summarized in more detail below.

ALEKS platform. The program provided access to ALEKS, a math-focused CAL software. ALEKS uses predictive algorithms to deliver personalized learning experiences through adaptive assessments and assignments. These tools evaluate students' knowledge by identifying the math topics they master, the ones they do not, and the ones they are ready to learn. This approach is grounded in Knowledge Space Theory (KST), which posits that a subject's knowledge can be mapped as a set of solvable problems (Doignon and Falmagne, 1985; Falmagne et al., 1990). Accordingly, ALEKS assesses students' mastery by their ability to solve representative problems from each topic.

The software encompasses over 600 mathematics topics for ninth graders. Given the infeasibility of covering all topics within an academic year, the curriculum was narrowed to grade-appropriate content and essential prerequisites, yielding 337 topics organized into nine modules corresponding to content areas scheduled for instruction during the academic year.⁴ Beyond practice exercises, ALEKS provides functionality for teachers to assign and evaluate homework and tests, along with a dashboard for tracking class and individual student progress.

All teachers assigned to a CAL classroom received an initial 2-hour training on the use of the learning technology. This training was provided by McGraw Hill, the developer of the learning platform. In addition to this, during the first 3 months of the program, teachers met one-on-one with a CAL aide for approximately one hour every two weeks. The CAL aide's role was to support them in accessing and monitoring students' assessments, tracking their progress, and identifying

²The study is registered as AEARCTR-0012383 in the AEA registry, and was approved under protocol number 24-444 by the Teachers College IRB.

³Figure 1 shows a full timeline of the study and intervention.

⁴This selection was performed by two local experts on math pedagogy based on a review of the curriculum. The majority of prerequisite topics were identified by the learning software.

topics that students had not yet mastered for lesson planning. Importantly, one of the primary aspects of this extended training was to help teachers utilize the software’s specific tools to ease their workload—including features that allow assigning and automatically grading homework—thus enabling teachers to save time that could potentially be devoted to other teaching activities.

Tutoring Intervention. The tutoring component explored complementarities between CAL and traditional instruction by providing up to three weekly hours of tutoring to academically at-risk students. In each classroom selected for tutoring, we targeted approximately 20 percent of students for the intervention. Sessions were managed by one or two tutors, depending on the number of tutees per classroom, which typically ranged from 5 to 8 students. To select tutoring recipients, mathematics teachers initially identified and ranked up to 10 students based on perceived academic need. To minimize selection bias, teachers were subsequently provided with baseline mathematics test results and given the opportunity to revise their selections; approximately 70 percent of teachers modified their initial selections after receiving this test score information, and we used the final teacher selections to determine tutoring assignments. Despite this safeguard, students ultimately selected for tutoring were not, on average, among the lowest baseline performers in their classrooms—a targeting gap we revisit in Section 6, where we consider its implications for the program’s effectiveness.

Tutors were selected through a competitive process that included interviews. All tutors were either recent graduates or in the final stages of earning a Bachelor’s degree in Mathematics with a focus on secondary education, and had experience in using information technology for teaching. Tutoring sessions were integrated into students’ schedules during elective workshop periods, covering approximately three academic hours per week, typically divided into one to three weekly sessions.⁵ Elective workshops are offered for all students attending extended-day schools in the Dominican Republic—schools where the day runs from 8am to 4pm and workshops typically cover areas outside of the core curriculum, such as computing or art.

To ensure quality and consistency, tutors were supported by a supervisor with training and experience in mathematics education for high school students. The supervisor reviewing individual session plans before each session, confirmed that topics and instructional methods had been prepared, and provided feedback to tutors as needed. Tutors also maintained close communication with the mathematics teacher to align their sessions with the classroom curriculum, and were instructed to incorporate the CAL platform into tutoring sessions.⁶ During the sessions, tutors reviewed topics covered in class during that week or revisited prior material, including prerequisite topics that students found challenging; for instance, if a class was working on linear equations, a

⁵Out of 9 tutoring classrooms, 2 sections had one weekly meeting of 3 hours, 5 sections had two weekly meetings of 2 and then 1 hour, and 2 sections had three meetings for 1 hour each day.

⁶For example, tutors were encouraged to use the platform to identify topics where students were lagging behind and review these during tutoring sessions, or use the platform’s accompanying exercises as practice. If a given session had a higher tutee/tutor ratio due to a tutor absence, the tutor was encouraged to alternate the use of CAL with small-group tutoring.

tutor might review the rules of signs before addressing the current topic.

The effectiveness of the tutoring sessions was monitored through unannounced visits, intended to identify potential issues or areas for improvement, followed by one-on-one feedback meetings between the supervisor and the observed tutor. Tutors also completed weekly surveys reporting the number of tutoring hours conducted for each class and student, to monitor attendance.

2.3 Sample and Randomization

The intervention was administered in 9 public high schools (38 ninth-grade classrooms) in the Dominican Republic. The sample size was constrained by schools’ ability to meet the required technical and operational conditions. Schools were selected based on four criteria: (1) having an active computer lab, (2) having regular access to electricity and internet, (3) having an extended-day format,⁷ and (4) having at least three ninth-grade classrooms. We further limited the sample to schools that had participated in *Programate*, a joint collaboration between MINERD and the World Bank that provided access to the ALEKS software to approximately 60 schools between 2018 and 2023. These criteria yielded an initial list of 34 schools. Between April and August 2023, the research team, with support from MINERD, contacted these schools by phone to verify eligibility assess interest. While all contacted schools expressed strong interest, only 14 confirmed reliable internet access and a functional computer lab.

Between September 4th and 15th, a member of the research team conducted school visits to validate the information collected during the spring and summer terms. Only 9 of the 14 visited schools had functioning internet and computers. All participating schools were located in Santo Domingo and Santiago—the two largest cities in the country and, together, home to approximately 45 percent of the national student population (Ministerio de Educación de la República Dominicana (MINERD), 2023)—so while the sample is geographically concentrated, it captures a substantial share of the national ninth-grade population.

Treatment was assigned using a two-stage randomization process at the classroom level. Classrooms were first ranked by size and paired with the closest matching classroom in terms of size; within each pair, one classroom was randomly assigned to the treatment group and the other to the control group, ensuring similarly sized classrooms across groups. This pairing and randomization process was then repeated within the treatment group to assign half of these classrooms to receive the additional tutoring component: new pairs were formed based on classroom size within the CAL group, and the tutoring intervention was randomly assigned within each pair. The study ultimately comprised 38 classrooms, with 19 assigned to receive the CAL software and the remaining 19 serving as controls.⁸ Of the 19 CAL classrooms, 9 were further selected to receive group tutoring. The

⁷Approximately 67 percent of public high schools in the Dominican Republic offered an extended school day format during the 2021-2022 school year (MINERD, 2023), where the school day starts at 8am and ends at 4pm.

⁸Students in control classrooms received temporary licenses to access the software approximately two months after endline data collection.

randomization design is summarized in Figure 2, and power calculations are included in Appendix A.1.

2.4 Implementation

The CAL intervention started on October 23rd and concluded on May 3rd.⁹ During the first week of implementation, all treated students took a mandatory “Initial Knowledge Check” through the learning platform, the first activity within ALEKS. Based on this initial assessment, the platform predicts the topics each student has mastered, has not yet mastered, and is ready to learn, and generates an individualized learning path accordingly.

The first tutoring session took place on October 23rd and the last on April 26th. Tutoring sessions took place one to three days per week during extracurricular hours. In all but one school, sessions took place during hours allocated for optional workshops.¹⁰ Optional workshops were meant to cover extracurricular content and varied in scope, including courses such as computing, music or dance. Workshops and tutoring sessions typically took place between 2:00 and 3:30pm. In total, 61 students were selected to participate in tutoring.

3 Data

To evaluate the effectiveness of the program, our primary outcome is students’ academic performance. We additionally examine a set of secondary outcomes—socioemotional skills, psychological well-being, and student-reported teacher behavior—to explore potential channels through which the interventions may have operated. These outcomes were collected using a combination of survey data, administrative records, data from the adaptive software, and brief math assessments, each described below.

Student surveys and math assessments were administered at baseline, midline, and endline by the research team. Baseline surveys were collected in September 2023, midline surveys in December 2023, and endline surveys in May 2024. Administrative records were collected and digitized after the end of the school year, between June and September 2024.

Academic performance. Our main outcome of interest is students’ academic performance, used to examine whether the CAL platform or tutoring component improved student learning in math. Assessments were designed using items of varying difficulty, adapted from MINERD’s national assessments. Endline tests were designed by a pedagogical expert and followed the content and cognitive areas used on the 2019 Trends in International Mathematics and Science Study (TIMSS) 8th-grade math assessments, covering seven content domains and three cognitive areas

⁹This is equivalent to roughly 21 weeks, excluding winter, Holy Week, and two weeks where the use of the platform was interrupted due to training and data collection.

¹⁰One of the nine schools did not offer optional workshops, so students stayed after school hours to receive tutoring sessions.

(Mullis and Martin, 2017).¹¹ We examine student performance across these content areas and cognitive domains at both at midline and endline, and classify items by difficulty level to assess differential impacts by grade-level content.

Secondary outcomes. Beyond academic performance, we collected several secondary measures to explore the channels through which the interventions may have affected students, though as discussed in Section 6, most of these outcomes show limited or no statistically significant treatment effects and should be interpreted as exploratory.

Administrative records. Between June and September 2024, we additionally collected and digitized schools’ administrative records. These records include students’ end-of-year grades in math and Spanish, attendance, dropout status, and grade promotion outcomes. End-of-year Spanish grades are included to test for adverse spillover effects from time reallocated away from other subjects.

Socioemotional skills. As students increasingly rely on digital devices for learning, there is a risk that socioemotional or non-cognitive skills may be negatively impacted. In baseline and endline surveys we measured the following three skills: perseverance, locus of control, and grit. Perseverance is measured through a real effort task adapted from Alan et al. (2019), in which students answer a logic question and then choose between attempting another question of the same difficulty, tackling a harder question, or giving up; their choice serves as our measure of perseverance. Locus of control—the extent to which students believe they have control over the outcome of events in their lives—is measured using an adaptation of Rotter’s locus of control scale (Rotter, 1966). Grit is measured using a 4-item scale adapted from Carlana and La Ferrara (2021). Grit and locus of control indices are normalized to range between 0 and 1.

Student well-being. Because ALEKS has each student work independently, it may offer fewer opportunities for peer interaction or support than regular instruction, potentially increasing feelings of isolation in the classroom. This concern is compounded by a broader literature linking increased screen time to lower psychological well-being among teenagers (Stiglic and Viner, 2019; Twenge and Campbell, 2018). To assess whether these dynamics affected our sample, we measure student well-being using items from the Children’s Depression Screener (ChilD-S) developed by Fröhe et al. (2012).

Increased use of CAL could plausibly affect math anxiety in either direction: improving students’ math skills, or their command of foundational learning, could reduce anxiety, while greater exposure to diagnostic feedback highlighting unmastered topics could increase it. Following Araya et al. (2019), we construct a math anxiety index from students’ Likert-scale responses to two statements: ”I get nervous before math tests” and ”I am worried of having bad grades in math.”

Teacher behavior. The CAL platform may free up teacher time by automating tasks such

¹¹The 8th grade TIMMS math assessments evaluated the following content domains: (1) Numbers, (2) Algebra, (3) Geometry, and (4) Data and Probability. The cognitive domains evaluated were: (1) Knowing, (2) Applying, and (3) Reasoning. TIMMS is only administered in the 4th and 8th grades.

as developing and grading homework and tests, potentially shifting how teachers allocate time and effort across classroom activities. To explore this, we ask students about their perceptions of teacher behavior (e.g., whether the teacher provided additional support outside class, or clarified a lesson that was not understood) and time management (e.g. e.g., whether the teacher started class late, or came to school). We use these responses to construct two indices: a time mismanagement index and a teacher engagement index, based on students’ reports of classroom engagement.¹² All indices are standardized to have a mean zero and a standard deviation of 1 in the control group. The questions used have been adapted from questionnaires used by De Hoyos et al. (2021) and Ferguson and Danielson (2015).

CAL data. Used for both monitoring and assessment purposes, the software records students’ interactions with the platform, including the number of sessions held, the date and duration of each session, and the number of newly mastered topics per student. We use this information to assess compliance with the intervention and to measure progress within the treatment group. This data is available for the treatment group only.

3.1 Balance Checks and Attrition

Table 1 presents balance checks using baseline survey data. There are no statistically significant differences between the treatment and control groups across observable characteristics, supporting the integrity of the randomization. The sample is approximately 56 percent female, with an average age of nearly 15—somewhat above the typical enrollment age of 13 to 14 for night graders, reflecting the high rates of grade repetition in the Dominican Republic (approximately 19 percent of students in our sample repeated at least one grade). Baseline academic performance is relatively low: students answered an average of 6 out of 12 questions correctly. The sample also skews toward lower socioeconomic backgrounds, with fewer than 40 percent of students having a parent who attended college.

Attrition was low throughout the study. As shown in Columns 1 and 5 of Table 2, 92 percent of the baseline sample completed surveys and assessments at midline, and 90 percent did so at endline. Completion rates were similar across treatment and control groups, with differences of 0.5 percentage points at midline and 2.9 percentage points at endline. Although the latter is statistically significant at the 5 percent level, its magnitude is small and unlikely to affect our estimates meaningfully.

More importantly, the students who completed the surveys were broadly similar to those who did not, as shown in Columns 2-4 and 6-8 of Table 2. A few minor differences in observable characteristics emerge: older students were slightly less likely to respond, with each additional

¹²The engagement index is built from student responses to: (1) Your math teacher keeps students engaged, (2) Asks questions to check for understanding, (3) Knows when everyone understands, (4) Makes every student work hard, (5) Assigns homework that helps you learn, (6) Reviews past topics, (7) Marks your work to help understanding, and (8) Wants you to explain your answers.

year of age reducing response probability by 3.7 percentage points; female students were about 4 percentage points more likely to complete surveys than males; and students with higher baseline test scores were marginally more likely to remain in the sample (0.9 percentage points per standard deviation). None of these differences are large enough to raise substantial concerns about sample composition. As a robustness check, Appendix Table A1, we present Lee (2009) bounds for the main performance outcomes, confirming that our findings are not driven by differential attrition.¹³

4 Empirical Framework

We estimate the impact of the program on student outcomes using the following regression:

$$Y_{ci} = \beta_0 + \beta_1 \text{CAL}_c + \beta_2 \text{Tutoring}_c + \beta_3 X_{ci} + \epsilon_{ci}$$

where Y_{ci} is the outcome of student i in classroom c and CAL_c is a binary variable equal to 1 if classroom c was assigned to the CAL treatment and 0 otherwise, and Tutoring_c is a binary variable equal to 1 if the classroom was additionally assigned to receive small-group tutoring and 0 otherwise. X_{ci} is a vector of baseline characteristics for student i in classroom c . Under this specification, β_1 captures the average treatment effect (ATE) of the CAL software relative to the control group, while β_2 captures the additional impact (“value-added”) of the tutoring component relative to classrooms that received CAL alone. In the results section, we report each coefficient separately as well as their sum, $\beta_1 + \beta_2$, which represents the total effect for classrooms assigned to both the CAL and tutoring interventions, relative to the control group.

Throughout this paper, we use randomization-based inference as our preferred estimation strategy, deriving statistical inferences directly from the randomization procedure rather than relying on assumptions about the sample distribution (Athey and Imbens, 2017; Green et al., 2011; Young, 2019). This approach is particularly well-suited to our setting for two reasons. First, it naturally accommodates the pairwise randomization structure used in this study, ensuring that inference respects the design under which treatment was assigned. Second, it yields nearly exact p-values without requiring large-sample approximations, which is especially valuable when the number of clusters is small, as is the case here.

The validity of our findings relies on the Stable Unit Treatment Value Assumption (Rubin, 1978), which requires that the treatment status of one classroom does not affect the potential outcomes of students in another. Several features of the study design support this assumption. Access to the ALEKS platform required each student to log in with a unique username and password, and platform use was closely monitored during designated sessions, making it unlikely

¹³We calculate bounds using a single regression with all three experimental groups ($y = \beta_1 \text{CAL} + \beta_2 \text{CAL} \times \text{Tutoring} + \epsilon$). All groups are trimmed to equalize response rates at the minimum observed level (89.8%). The lower bound drops the highest-scoring students from higher-response groups (CAL and CAL+Tutoring); the upper bound drops the lowest-scoring students, ensuring treatment effect estimates are not biased by differential attrition.

that students in control classrooms could have accessed or benefited from the software. To assess the possibility of indirect spillovers—through treated and control students interacting outside of class, or through teachers adjusting their behavior in non-treated classrooms—we conducted discussions with the teaching team and focus groups with students. Neither revealed evidence that teachers modified instruction in control classrooms in response to the treatment, nor that treated students collaborated with control peers on academic tasks. While this does not definitively rule out interference, we find no evidence that treatment affected the behavior of students or teachers in control classrooms.

5 Results

5.1 CAL and Tutoring Take-Up

Results from the initial ALEKS assessment reveal that, at the time of completion, the median student mastered only 2.7 percent (9 out of 337) of the math topics included in the course, confirming the substantial learning deficits documented in the background section. Over the course of the school year, the median treated student spent approximately 14 hours on the platform (approx. 38 minutes per week) and learned an average of 45 new math topics. As shown in Figures 3 and 4, both distributions are left-skewed: students at the 90th percentile spent over 38 hours on the platform and learned more than 188 new math topics by the end of the year.

To assess out-of-school usage, we measure the share of total platform time occurring on non-scheduled days. The median student used the platform outside school hours for approximately two hours, representing 15 percent of their total platform time. This modest figure suggests that the effects we document below reflect a reallocation of in-class instructional time rather than an increase in total learning time.

Take-up of the tutoring component was lower than intended. Of the 61 students assigned to receive tutoring, all received at least 2 hours of sessions, with the median student receiving 29 hours over the school year—approximately 1.4 academic hours per week, well below the intended dosage of 3 hours per week. Attendance was a primary driver of this shortfall: on any given week, up to 40 percent of students failed to show up to at least one scheduled session, based on monitoring data. We discuss potential explanations for this compliance gap in Section 6.

5.2 Student Performance

Table 3 presents Intention-to-Treat (ITT) estimates approximately two months (midline) and six months (endline) after the start of the program. Columns 1 and 2 report midline results and Columns 3 and 4 report endline results, with and without baseline covariates.

At midline, CAL-only classrooms showed gains of 0.17-0.21 standard deviations relative to control, as shown in Columns 1 and 2. While economically meaningful, these effects are not

statistically significant, consistent with the program having had limited time to generate detectable impacts at this stage. These impacts are roughly consistent across specifications (with or without covariates). We similarly find no significant effects of the tutoring component at midline, which is unsurprising given that students had received only 8 tutoring sessions by that point.

By endline, the CAL-only effect had increased to 0.28-0.31 standard deviations as shown in Columns 3 and 4. These gains are consistent with the broader literature on computer-adaptive learning in low- and middle-income countries: Araya et al. (2019) report effects of 0.21-0.27 SD from a game-based math platform in Chile after seven months; Muralidharan et al. (2019) find effects of 0.37 SD in India after 4.5 months; and Büchel et al. (2022) document effects of up to 0.17 SD from a computer-assisted learning software in El Salvador over six months. Our estimates fall within this range and are notable in that they are achieved through direct substitution of regular classroom instruction, without additional instructional time or dedicated implementation staff.

The tutoring component, by contrast, did not enhance these gains. The coefficient on the tutoring add-on (β_2) is negative (-0.19 SD with covariates), though not statistically distinguishable from zero. The combined effect of CAL plus tutoring relative to control ($\beta_1 + \beta_2 = 0.09$ SD) is also not statistically significant, implying that students in classrooms assigned to both interventions performed no better than the control group in average. While we cannot reject the null hypothesis that tutoring had zero effect, the direction and magnitude of the point estimates are consistent with the two interventions working against each other rather than reinforcing one another. In line with this interpretation, students in tutoring classrooms—including those not directly receiving tutoring—exhibited reduced engagement with the CAL platform (Figure 9) and lower locus of control (Appendix Table B8) relative to the CAL-only classrooms, suggesting that the introduction of tutoring may have altered student behavior in ways that eroded the effectiveness of the adaptive technology. We explore these mechanisms in detail in Section 6.

5.3 Heterogeneity in Program Effects

We explore heterogeneity in CAL’s effects along three dimensions: content difficulty, mathematical domain, and cognitive skill area.

Following Muralidharan et al. (2019), we first classify test items into two categories—below and at grade level—and test whether students fare differentially benefited students depending on the complexity of the material. Table 4 shows that the CAL treatment improved performance on both below and at grade-level content, with effects ranging from 0.21 to 0.28 standard deviations, though the at grade-level effect loses statistical significance without covariates. This pattern differs from Muralidharan et al. (2019), who find large impacts on below grade-level items but no impacts on at grade-level content, as is consistent with the structure of the ALEKS platform, which was designed to cover both difficulty levels.¹⁴ Supporting this interpretation, midline results—when

¹⁴In Hindi, the authors find significant effects for questions at and below grade level.

students had spent the majority of their time on modules covering below grade-level topics—show effects concentrated almost entirely on below-grade test items, with at grade-level effects emerging only by endline as students progressed through the curriculum. Table 5 present effects across the four content areas evaluated in the TIMMS 8th-grade mathematics assessment. By endline, we find positive effects of the CAL treatment on the algebra, numbers, and geometry, though the geometry coefficient is not statistically significant. We find no effect on data and probability, the one TIMMS content area that was not covered by the platform or included in the the Dominican Republic’s 9th-grade curriculum—a pattern that supports the interpretation that observed gains reflect learning through the platform rather than general test-taking improvements. Results for the tutoring component mirror those in Table 3: classrooms assigned to tutoring show negative but statistically insignificant differences relative CAL-only classrooms across all content areas except data and probability.

Table 6 presents results across the three cognitive domains evaluates by TIMMS: knowing, applying, and reasoning. We find large positive effects on all three domains at endline, ranging from 0.16 to 0.31 standard deviations, though the effect on the applying domain does not reach statistical significance.

Lastly, Figure 6 plots the distribution of endline performance for both the treatment and control groups across baseline performance percentiles. Despite overlapping confidence intervals at several points, the treatment group consistently outperforms the control group across the full distribution of baseline performance, suggesting that the program’s benefits were not concentrated among higher- or lower-performing students.

5.4 Secondary Outcomes

Beyond academic performance, we examine whether the interventions affected student retention, motivation, socioemotional skills, psychological well-being, beliefs about own performance, and teacher behavior. Most secondary outcomes show no statistically significant effects. The notable exceptions are: greater overconfidence in anticipated test performance and marginally significant reductions in dropout rates and locus of control among CAL+Tutoring students; and increased preference for easier tasks on a real-effort persistence measure under CAL-only students. We summarize these findings briefly here; full results are reported in Appendix B.

Student Retention and Motivation. Despite the substantial learning gains from the CAL intervention, we find no effects on dropout rates, school absenteeism, intrinsic motivation, or stated preferences for mathematics among CAL-only students. The null effect on dropout is consistent with students facing binding non-academic constraints—such as economic pressures or family obligations—that learning gains alone cannot address. The CAL+Tutoring group shows a marginally significant reduction in dropout rates at endline. Given the absence of academic

performance gains in this arm, this effect is unlikely to operate through improved learning; instead, it may reflect the added benefit of regular human contact and personalized attention from a tutor.

Student Beliefs and Expectations. We examine whether the interventions affected students’ calibration of their own academic performance. At endline, students were asked to predict their test score prior to taking the assessment, allowing us to construct a measure of overconfidence as the difference between predicted and actual performance. As shown in Table 7, students in CAL+Tutoring classrooms exhibited marginally higher overconfidence than the control group (5.9 percentage points, $p < 0.10$ with covariates), while CAL-only students showed no significant difference. Figure 5 illustrates these patterns: while actual performance was highest in CAL-only classrooms, expected performance was similar across all three groups, resulting in greater overconfidence among CAL+Tutoring students—whose actual gains were smaller—rather than among those who benefited from the intervention. We discuss the implications of this belief miscalibration for platform engagement in Subsection 6.4.

Socioemotional Skills. We find no significant effects of either intervention on self-efficacy or grit at endline. Students in CAL-only classrooms were, however, more likely to choose an easier task over a more challenging alternative in the real-effort perseverance exercise, suggesting that sustained exposure to an adaptive platform—which continuously adjusts difficulty to each student’s level—may reduce willingness to voluntarily tackle harder problems. For locus of control, the tutoring component produced a marginally significant negative effect at endline ($p < 0.10$), indicating that students in the CAL+tutoring group were more likely to attribute outcomes to external factors rather than their own effort—a finding consistent with the broader pattern of reduced student agency we document in Section 6.

Psychological Well-Being. We find no significant effects on a depression index based on the Children’s Depression Screener, nor on the overall math anxiety index at endline. Students in CAL-only classrooms showed a marginally significant increase in nervousness before tests relative to control, while the combined effect for CAL+Tutoring classrooms was indistinguishable from zero. Rather than reflecting distress, this pattern may indicate that students with greater exposure to the platform’s diagnostic feedback developed heightened awareness of their own knowledge gaps—consistent with the higher platform engagement we document for CAL-only students relative to their CAL+Tutoring peers.

Student Time Use. One potential concern is that CAL might crowd out other productive activities or increase screen time outside of school. We find no significant effects on students’ time allocation across eight categories: in-person classes, unpaid work at home, unpaid work outside the home, paid work, studying outside of school, meeting friends, entertainment, and sports. This

suggests that the intervention operated primarily through the reallocation of in-class instructional time rather than through changes in how students allocated time outside of school.

Teacher Behavior. We find no significant effects on either the teacher time mismanagement index or the teacher engagement index at endline. The absence of detectable changes in teacher behavior suggests that the integration of the CAL platform into regular instruction did not substantially alter how teachers allocated time or engaged with students during non-platform hours, and that the learning gains we document are more likely attributable to the direct substitution of adaptive instruction for traditional class time than to changes in teacher practice

6 Mechanisms

The results indicate that the impact of the CAL intervention depends critically on students' engagement with the platform and on how the intervention interacts with existing instructional practices. In this section, we first rule out differential implementation as an explanation for the divergent effects between treatment arms, and then synthesize quantitative evidence from platform usage, student beliefs, and performance outcomes, together with qualitative findings from focus groups, to discuss potential mechanisms.

6.1 Implementation Fidelity

To assess whether differential implementation could explain the divergent effects, we collected weekly monitoring data on session delivery across all 19 CAL classrooms over 21 weeks.¹⁵ Overall, CAL sessions achieved 67.7 percent of expected dosage (28.4 of 42 planned sessions). Fidelity did not differ between classrooms assigned to CAL only versus CAL plus tutoring: the former achieved 68.8 percent compliance while the latter achieved 66.4 percent. Among the nine classrooms receiving both components, tutoring achieved 63.5 percent of expected dosage—statistically indistinguishable from CAL compliance in those same classrooms.

When sessions were cancelled, the reasons were predominantly exogenous. Among cancelled CAL sessions, 74.5 percent were due to school closures, institutional activities, or infrastructure failures; similarly, 75.5 percent of cancelled tutoring sessions were attributable to such factors. The remaining cancellations reflected intervention-specific barriers: laboratory access and ALEKS licensing issues for CAL, and tutor availability and space constraints for tutoring. The similar compliance rates and barrier distributions across components rule out differential implementation as an explanation for the attenuated effects in the CAL plus tutoring arm.

¹⁵Compliance rates are calculated at the classroom level (sessions held divided by sessions scheduled).

6.2 Engagement Intensity and Dose–Response

A key mechanism through which the CAL intervention appears to affect learning is sustained engagement with adaptive, self-directed practice. Platform data reveal substantial heterogeneity in usage: the median treated student used ALEKS for approximately 14 hours over the school year and mastered 45 topics, while students at the 90th percentile spent over 38 hours on the platform and mastered more than 188 topics (see Figures 3 and 4). This variation is economically meaningful. Using random assignment as an instrument for hours of platform use, we estimate that each additional hour of CAL engagement increases endline test scores by approximately 0.026–0.027 standard deviations (see Table 8).

The estimated dose–response relationship exhibits diminishing returns, with gains increasing steeply up to roughly 1500 minutes (25 hours) of use before flattening (see Figure A2). In Table 8 we extrapolate our estimates to 1500 minutes of use—equivalent to roughly 5 months of implementation at 80 percent attendance—suggests that fully faithful implementation could generate test score gains of 0.73 standard deviations.¹⁶ While such extrapolation should be interpreted cautiously, these estimates suggest that implementation frictions—including connectivity issues, limited device availability, and school disruptions—reduced effective exposure to the platform and may have attenuated impacts relative to this benchmark.

6.3 Substitution of Effort and Classroom-Level Spillovers

Despite similar access to the platform, students in CAL+tutoring classrooms spent less time on ALEKS and mastered fewer topics than students in CAL-only classrooms. Students in the tutoring arm also exerted less effort while using the platform.¹⁷ Importantly, this pattern is observed not only for students selected for tutoring, but also for their non-tutored classmates, indicating that the tutoring intervention affected classroom-level learning dynamics rather than only individual tutees.

Figure 9 shows that the average student in a CAL-only classroom spent a total of 1,232 minutes using the platform (56 minutes per week), compared to 1,082 minutes (49 minutes per week) for the average student in a CAL+Tutoring classroom. Using the estimated dose-response relationship, this difference in time usage accounts for approximately 34 percent (0.07 SD) of the observed gap in endline performance between the two treatment arms (see Table 8). Within CAL+Tutoring classrooms, both tutees and non-tutees used the platform less than their CAL-only classroom, though the gap was larger for tutees, who averaged only 1,000 minutes (or 45 minutes per week).

One interpretation is that tutoring altered how students allocated learning effort. Qualitative evidence suggests that students in tutoring classrooms often relied on tutors to resolve difficulties rather than persisting independently. Although tutoring dosage fell short of intended levels due to

¹⁶This extrapolation requires the assumption of heterogeneity in treatment effects. As shown in Figure 6, we do not find evidence of effect heterogeneity across baseline performance.

¹⁷We define effort as the ratio of topics learned to time spent on the platform.

absenteeism and scheduling disruptions, the availability of a tutor may nonetheless have provided an alternative pathway to understanding that substituted for self-directed platform engagement.

6.4 Agency, Beliefs, and Attribution

Differences in engagement across treatment arms are accompanied by differences in students' beliefs about their own learning. Survey data indicate that students in CAL-only classrooms formed more accurate expectations about their academic performance, while students in CAL+tutoring classrooms exhibited higher overconfidence and a marginally significant reduction in locus of control, indicating that they were more likely to attribute outcomes to external factors rather than their own effort (see Tables 7 and B8). These patterns suggest that the presence of a tutor may have shifted students' attribution of learning away from individual effort and toward external support.

Platform design features—such as difficulty jumps, loss of progress after errors, and limited explanatory feedback—likely increased the cost of persistence for some students, as highlighted in focus group discussions. In CAL-only classrooms, students had no alternative source of support and may therefore have been more likely to persevere through those frictions. In tutoring classrooms, access to human explanations may have reduced tolerance for platform-related challenges, reinforcing both disengagement from the software and a greater tendency to attribute learning outcomes to the tutor rather than to one's own effort. While these mechanisms cannot be conclusively established from the available data, they are consistent with the observed patterns in platform engagement, beliefs, and performance.

6.5 Implications

Taken together, the evidence suggests that the absence of additional gains from the tutoring add-on reflects more than implementation shortfalls. Rather, tutoring appears to have interacted with the CAL intervention in ways that weakened the mechanisms through which the technology generates learning gains—by reducing platform engagement, lowering effort on the platform, and shifting students' beliefs about the source of their learning. These behavioral responses were not confined to tutees but extended to their non-tutored classmates, suggesting that the presence of tutoring reshaped classroom-level incentives and norms around self-directed learning.

These findings highlight the importance of considering behavioral responses when combining educational interventions. Adaptive learning technologies may be most effective when they preserve and reinforce student agency and sustained self-directed engagement. When layered with interventions that substitute for these processes—particularly in settings with implementation frictions—their effectiveness may be diminished. More broadly, the results underscore the value of empirically testing intervention bundles rather than assuming their effects are additive.

7 Conclusion

This paper provides experimental evidence on the effectiveness of computer-adaptive learning as a substitute for regular math instruction, and on whether combining it with small-group tutoring generates complementary gains. Conducted in the Dominican Republic—one of the lowest-performing countries in international assessments—our results speak to both the potential and the limits of technology-based approaches in resource-constrained settings.

We find that replacing two of seven weekly math hours with an adaptive learning platform improved student performance 0.28-0.31 standard deviations after 6 months, with effects robust across specifications, mathematical domains and difficulty levels. These gains place our estimates within the upper range of effects documented for computer-adaptive learning in low- and middle-income countries, and are notable in that they are achieved through direct substitution of regular classroom instruction, without additional instructional time or dedicated implementation staff.

Adding small-group tutoring did not enhance these gains. Point estimates suggest that classrooms assigned to both interventions performed no better than the control group on average, and worse than CAL-only classrooms, though the latter difference is not statistically significant. The evidence points to behavioral mechanisms: students in tutoring classrooms—including those not directly tutored—exhibited reduced platform engagement, lower effort on the platform, and lower locus of control relative to CAL-only students. These patterns suggest that the availability of human support altered how students allocated learning effort, substituting tutor-provided explanations for the self-directed persistence that drives adaptive technology’s effectiveness. Reduced platform engagement accounts for approximately 34 percent of the observed performance gap between the two treatment arms, with the belief and attribution shifts we document suggesting additional behavioral channels, though their precise contribution to the remaining gap cannot be quantified from the available data.

Several broader lessons emerge from these findings. First, the effectiveness of adaptive learning technology appears to depend critically on the quality and intensity of student engagement, not merely on access. Implementation frictions that reduce time on the platform—whether from infrastructure constraints or behavioral responses to co-interventions—can substantially attenuate impact. Second, combining effective interventions does not guarantee additive benefits. When two components interact in ways that undermine one another’s mechanisms, the resulting bundle may perform worse than either component alone. Third, the classroom-level spillovers we document—where non-tutored students in tutoring classrooms also reduced their platform engagement—suggest that intervention design must account for how components shape broader classroom norms and incentives, not only the behavior of directly targeted students.

As artificial intelligence and adaptive technologies become increasingly integrated into education systems worldwide, understanding how to effectively combine human and technological inputs will grow in importance. This study provides evidence that such integration requires careful attention

to how components interact at the behavioral level, and that empirical testing of intervention bundles—rather than the assumption of additive effects—is essential for effective policy design.

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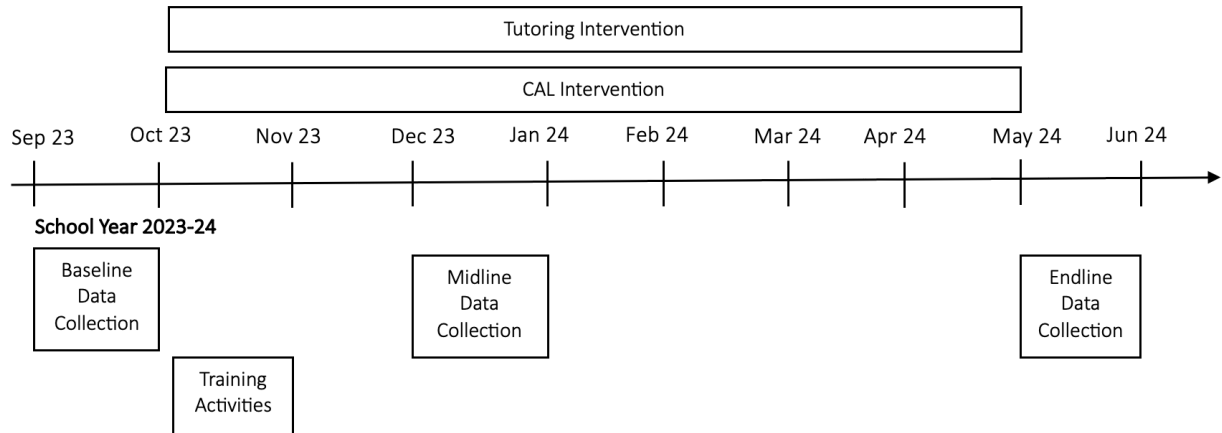
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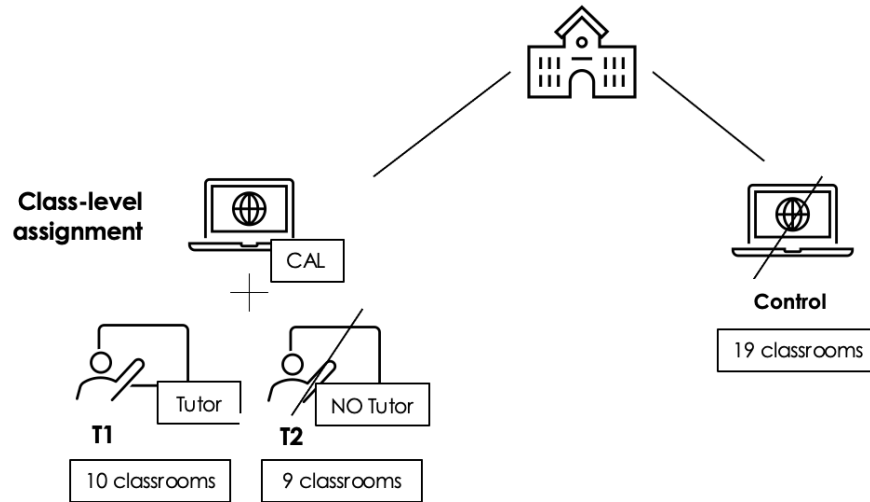
Figures and Tables

Figure 1: Timeline for Intervention and Study



Notes: The CAL intervention ran from October 23, 2023 to May 3, 2024 (approximately 21 instructional weeks). The tutoring intervention ran from October 23, 2023 to April 26, 2024. Platform use was interrupted during winter break, Holy Week, and two weeks allocated for training and data collection. Baseline surveys were administered in September 2023, midline surveys in December 2023, and endline surveys in May 2024

Figure 2: Randomization Design



Notes: T2 is composed of 10 classrooms assigned to receive access to the CAL software. T1 is composed of 9 classrooms where in addition to receiving access to the software, the bottom quarter of students, in terms of performance, were to receive tutoring sessions in small groups during extracurricular hours. The control group, containing 19 classrooms, did not receive access to the CAL or tutoring components.

Figure 3: Time Spent on CAL

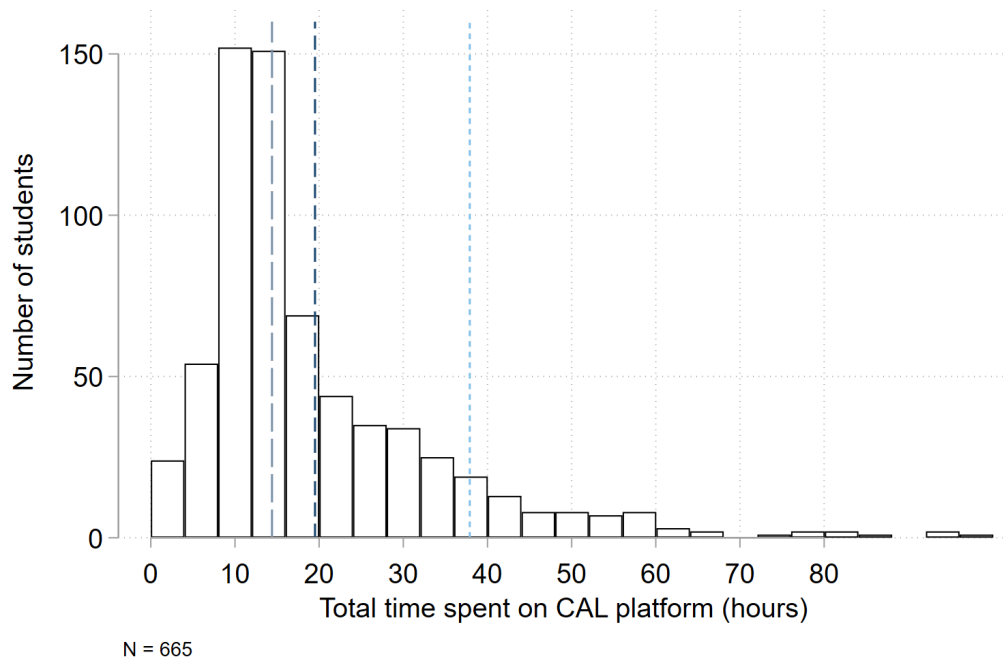


Figure 4: Topics Learned on CAL

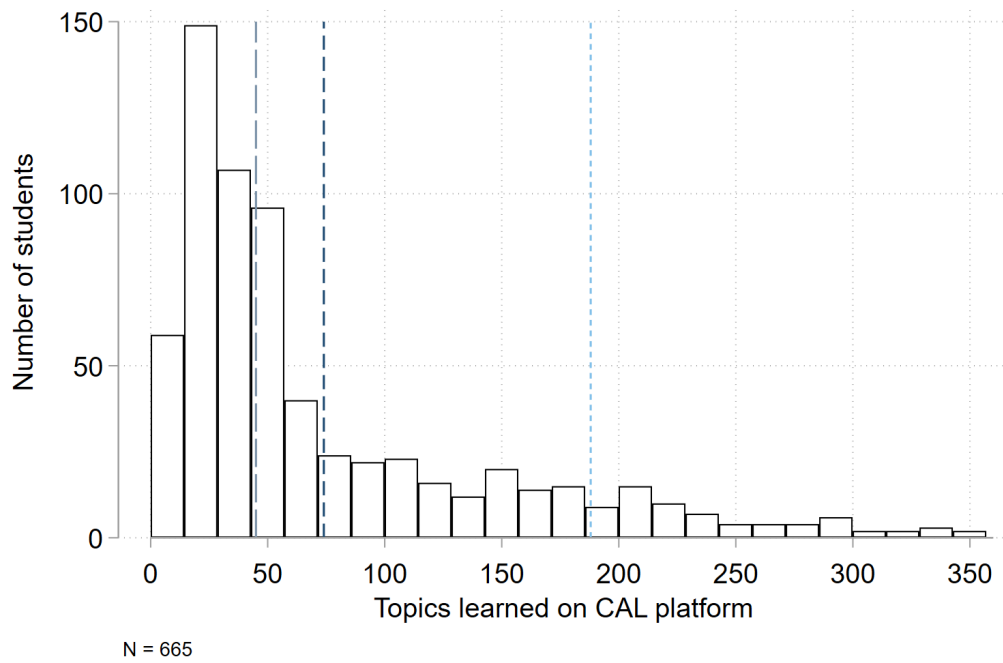
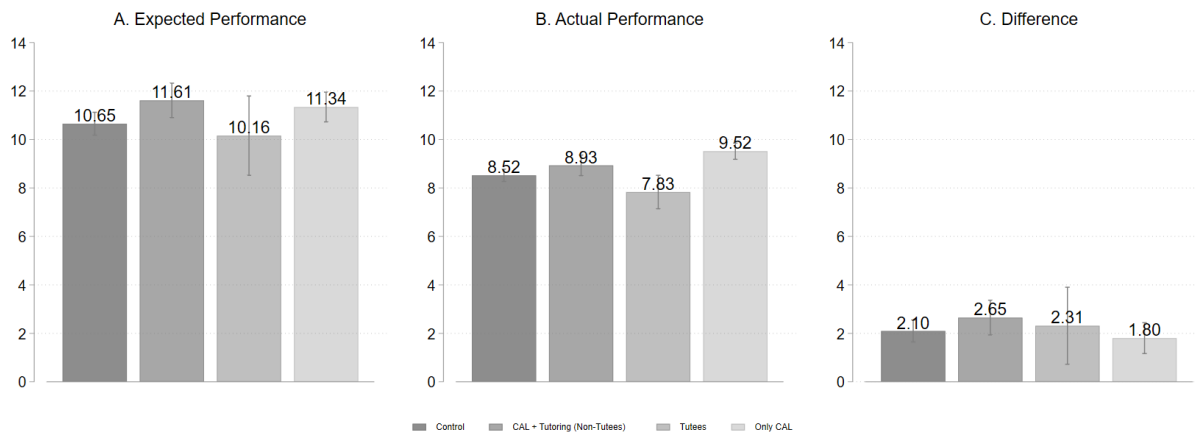
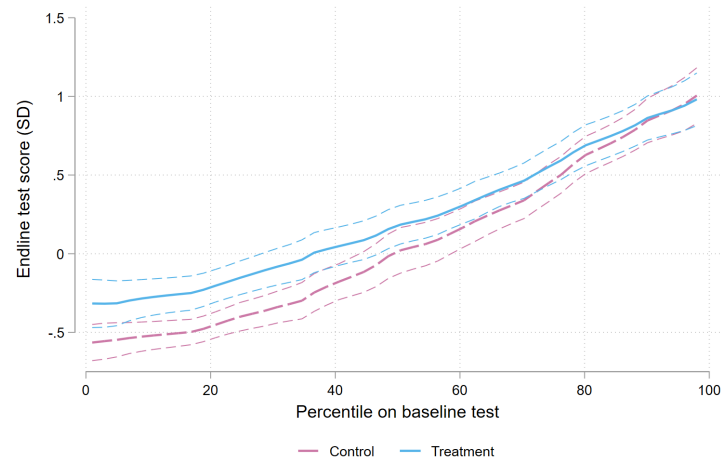


Figure 5: Expected Performance on Endline Math Tests



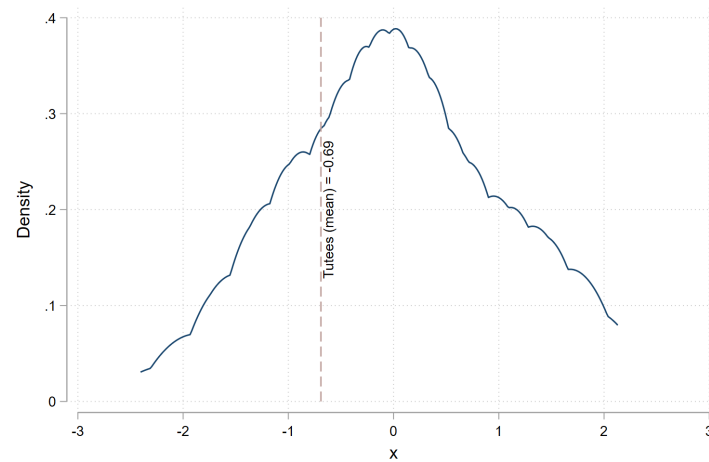
Notes: The CAL+Tutoring group excludes tutees. Panel C shows the difference between students' reported expected performance (share of correct answers) and their actual performance.

Figure 6: Non-Parametric Estimates of Treatment Effects by Baseline Performance



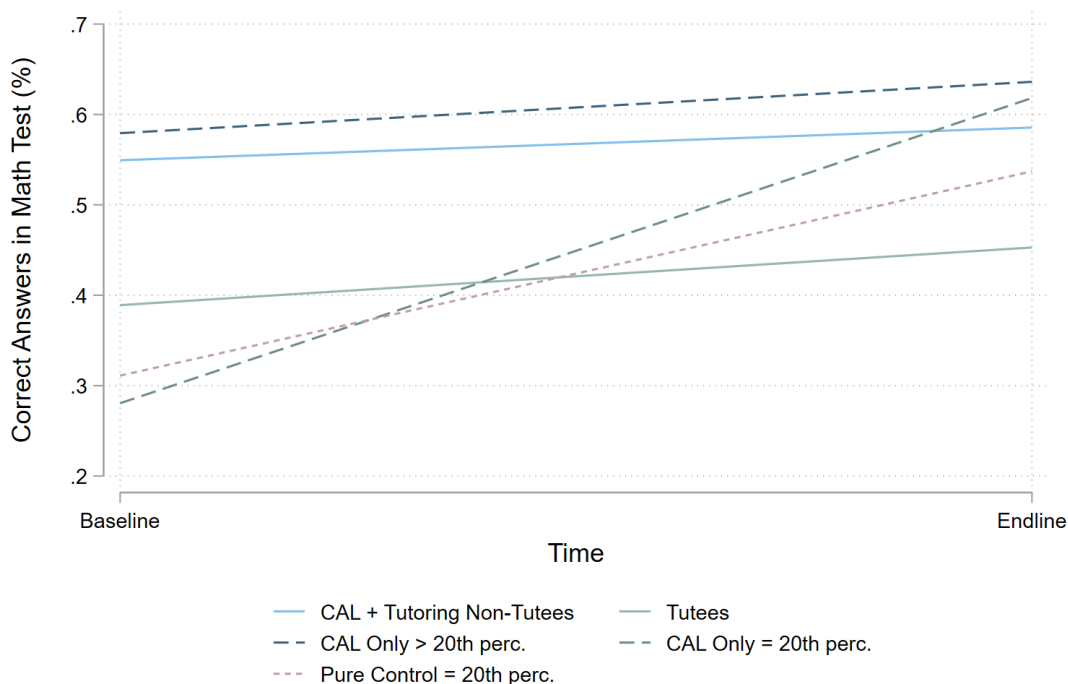
Notes: Following Muralidharan et al. (2019), the figure presents kernel-weighted local mean smoothed plots relating endline test scores to percentiles in baseline tests, separately for the treatment and control groups. The dashed lines show 95% confidence intervals.

Figure 7: Selection of Tutees and Distribution of Performance



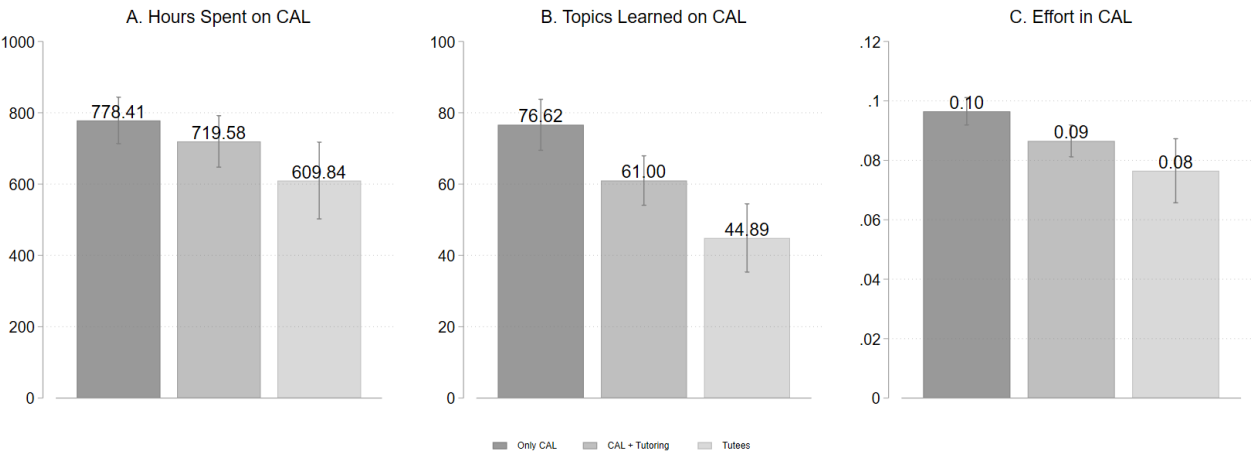
Notes: The blue line shows the average baseline score for selected tutees.

Figure 8: Share of Correct Answers by Performance Groups



Notes: The 20th percentile or below threshold of performance is used as a counterfactual for tutees. “CAL+Tutoring Non-Tutees” shows trends between baseline and endline for students who did not directly receive the group tutoring intervention but were in classrooms where approx. 20 percent of students did. “Tutees” are students who were assigned to the tutoring intervention in selected classrooms. “CAL-Only > 20th perc.” are students in CAL-only classrooms who are above the 20th percentile of performance. “CAL Only ≤ 20th percentile” are students in CAL-only classrooms who were below the 20th percentile of performance. “Pure Control ≤ 20th percentile” are students below the 20th percentile performance threshold in control classrooms. Dashed lines are approximate counterfactual trends for students in group tutoring classrooms who did not receive tutoring (blue) and students in tutoring classrooms who did receive tutoring (green and pink).

Figure 9: Platform Use by Groups



Notes: The CAL+Tutoring group excludes tutees.

Table 1: Balance Table

Variable	Control		Treatment		N	Difference
	N	Mean/(SE)	N	Mean/(SE)		
Female	671	0.553 (0.019)	662	0.565 (0.019)	1333	-0.012
Age	617	14.524 (0.033)	627	14.459 (0.033)	1244	0.064
Math test scores	622	6.280 (0.109)	629	6.444 (0.110)	1251	-0.164
Mother went to college	624	0.295 (0.018)	634	0.317 (0.018)	1258	-0.022
Father went to college	624	0.188 (0.016)	634	0.211 (0.016)	1258	-0.024
Mother went to high school	624	0.692 (0.018)	634	0.708 (0.018)	1258	-0.016
Father went to high school	624	0.566 (0.020)	634	0.557 (0.020)	1258	0.009
F-test of joint significance						0.652
F-test, number of observations						1237

Notes: *** p<0.01, ** p<0.05, * p<0.1. Math test scores are based on a short assessment administered at baseline.

Table 2: Attrition between Baseline, Midline, and Endline

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Midline Survey				Endline Survey			
	All	All	Treat	Control	All	All	Treat	Control
Treatment	0.005 (0.015)	0.013 (0.014)			0.029* (0.015)	0.025 (0.016)		
Student's age		-0.009 (0.011)	-0.022 (0.014)	0.007 (0.017)		-0.025** (0.012)	-0.034** (0.015)	-0.018 (0.020)
Student is female		-0.028* (0.015)	-0.020 (0.021)	-0.034 (0.023)		0.008 (0.017)	0.036 (0.022)	-0.013 (0.026)
Father is unemployed		-0.024 (0.027)	-0.036 (0.038)	-0.002 (0.039)		-0.002 (0.030)	0.036 (0.040)	-0.027 (0.045)
Mother is unemployed		-0.016 (0.017)	0.012 (0.023)	-0.046* (0.024)		0.001 (0.019)	0.019 (0.025)	-0.013 (0.028)
Baseline test score		0.003 (0.003)	0.002 (0.004)	0.005 (0.004)		0.008** (0.003)	0.011** (0.004)	0.005 (0.005)
Repeated a school year		-0.011 (0.024)	0.024 (0.033)	-0.043 (0.036)		-0.005 (0.027)	0.036 (0.035)	-0.038 (0.042)
Mother went to college		0.020 (0.018)	0.023 (0.024)	0.019 (0.027)		-0.012 (0.020)	-0.018 (0.026)	-0.005 (0.031)
Mother went to high school		0.031* (0.019)	0.014 (0.026)	0.044 (0.028)		0.027 (0.021)	0.014 (0.028)	0.039 (0.032)
Father went to high school		-0.032* (0.017)	0.013 (0.024)	-0.072*** (0.025)		-0.024 (0.019)	-0.030 (0.026)	-0.018 (0.029)
Father went to college		-0.006 (0.020)	-0.010 (0.026)	-0.012 (0.032)		-0.010 (0.022)	0.003 (0.028)	-0.017 (0.036)
Locus of control		0.058 (0.062)	0.095 (0.085)	0.021 (0.091)		0.001 (0.069)	0.018 (0.091)	-0.001 (0.105)
Self-efficacy		-0.031 (0.047)	-0.056 (0.064)	-0.011 (0.069)		-0.056 (0.053)	-0.081 (0.069)	-0.040 (0.080)
Spanish grade in 2022/23		0.003 (0.008)	0.002 (0.011)	0.006 (0.012)		0.000 (0.009)	0.003 (0.012)	-0.003 (0.014)
Math grade in 2022/23		-0.010 (0.009)	-0.005 (0.012)	-0.018 (0.014)		0.008 (0.010)	0.013 (0.013)	0.000 (0.016)
Constant	0.917*** (0.011)	1.075*** (0.168)	1.227*** (0.218)	0.886*** (0.263)	0.900*** (0.011)	1.240*** (0.187)	1.343*** (0.234)	1.184*** (0.303)
R-squared	0.000	0.025	0.027	0.048	0.003	0.023	0.038	0.022
N	1333	991	492	499	1333	991	492	499

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.

Table 3: Treatment Effects on Math Assessments

	Midline		Endline	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.2067 (0.1372)	0.1670 (0.1143)	0.3144* (0.1668)	0.2860** (0.1431)
CAL $\times$ Tutoring (β_2)	-0.1133 (0.1578)	-0.0770 (0.1251)	-0.2486 (0.1708)	-0.2045 (0.1454)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0934 (0.1346)	0.0900 (0.1142)	0.0658 (0.1486)	0.0815 (0.1268)
N	1225	1225	1218	1218
Adjusted R ²	0.005	0.277	0.016	0.302
Covariates	No	Yes	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ based on RI p-values.

Table 4: Treatment Effects on Test Scores by Content Level

	Below Grade Level		At Grade Level	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.2846* (0.1463)	0.2648** (0.1324)	0.2400 (0.1445)	0.2106* (0.1193)
CAL $\times$ Tutoring (β_2)	-0.1747 (0.1514)	-0.1405 (0.1397)	-0.2623 (0.1505)	-0.2204* (0.1192)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.1099 (0.1296)	0.1243 (0.1162)	-0.0223 (0.1302)	-0.0098 (0.1080)
N	1218	1218	1217	1217
Adjusted R ²	0.013	0.218	0.010	0.239
Covariates	No	Yes	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ based on RI p-values.

Table 5: Impact on Test Scores by Math Area

	Midline			Endline			
	Algebra (1)	Numbers (2)	Data & Prob (3)	Algebra (4)	Numbers (5)	Data & Prob (6)	Geometry (7)
CAL Treatment (β_1)	0.190 (0.112)	0.101 (0.110)	-0.021 (0.061)	0.211* (0.119)	0.325** (0.146)	0.027 (0.089)	0.130 (0.113)
CAL $\times$ Tutoring (β_2)	-0.102 (0.116)	0.004 (0.126)	-0.062 (0.078)	-0.220* (0.119)	-0.115 (0.152)	0.035 (0.098)	-0.192 (0.126)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.088 (0.109)	0.105 (0.105)	-0.083 (0.067)	-0.010 (0.108)	0.210 (0.130)	0.062 (0.082)	-0.061 (0.105)
N	1225	1225	1220	1217	1217	1218	1218
Adjusted R ²	0.189	0.187	0.055	0.239	0.180	0.044	0.080

Notes: All specifications include covariates. Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ based on RI p-values.

Table 6: Treatment Effects on Test Scores by Knowledge Domain

	Midline			Endline		
	Knowing (1)	Applying (2)	Reasoning (3)	Knowing (4)	Applying (5)	Reasoning (6)
CAL Treatment (β_1)	0.201 (0.140)	0.067 (0.087)	0.086 (0.079)	0.320*** (0.121)	0.158 (0.127)	0.208* (0.107)
CAL $\times$ Tutoring (β_2)	-0.012 (0.146)	-0.062 (0.093)	-0.156* (0.082)	-0.225* (0.135)	-0.104 (0.132)	-0.090 (0.105)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.189 (0.136)	0.004 (0.086)	-0.070 (0.072)	0.095 (0.111)	0.054 (0.113)	0.118 (0.093)
N	1225	1225	1224	1218	1217	1218
Adjusted R ²	0.178	0.180	0.076	0.186	0.173	0.101

Notes: All specifications include covariates. Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ based on RI p-values.

Table 7: Treatment Effects on Math Overconfidence

	Midline		Endline	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.0217 (0.0446)	0.0359 (0.0444)	-0.0032 (0.0338)	0.0070 (0.0353)
CAL $\times$ Tutoring (β_2)	0.0264 (0.0406)	0.0275 (0.0424)	0.0456 (0.0347)	0.0522 (0.0348)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0481 (0.0427)	0.0634 (0.0419)	0.0423 (0.0332)	0.0592* (0.0347)
N	1205	1205	1188	1188
Control Mean	0.560	0.560	0.580	0.580
Adjusted R ²	-0.000	0.051	-0.000	0.055
Covariates	No	Yes	No	Yes

Notes: The outcome variable takes the value of 1 if the teacher switched classes after program assignment, and 0 otherwise. Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table 8: Dose-response of ALEKS Use

	Standardized Math Scores		
	IV estimates (1)	OLS VA (full sample) (2)	OLS VA (treatment group) (3)
CAL hours	0.0269*** (0.0057)	0.0188*** (0.0031)	0.0254*** (0.0051)
Baseline math score (SD)	0.1433*** (0.0100)	0.1459*** (0.0099)	0.1129*** (0.0144)
Observations	1218	1218	615
R-squared	0.3229	0.3286	0.3174
Difference-in-Sargan statistic for exogeneity (p-value)	0.0882		
Extrapolated estimates of 2.5-hour treatment	0.0673	0.0469	0.0635
Extrapolated estimates 25 hours (1500 minutes)	0.6731	0.4689	0.6353

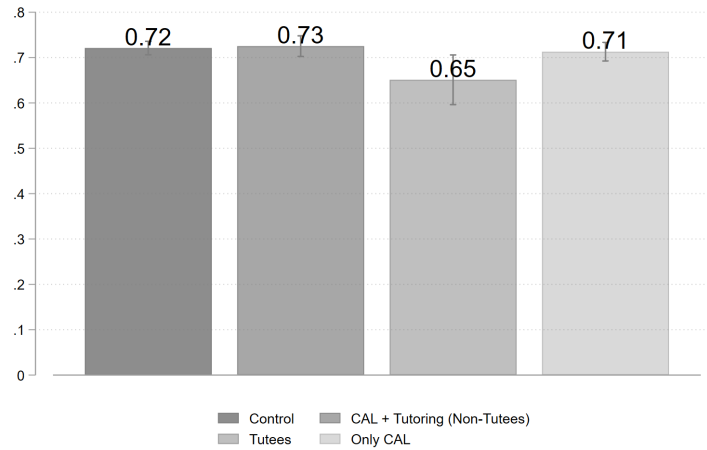
Notes: Robust standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01. Treatment group students who were randomly-selected for the ALEKS platform offer but who did not take up the offer have been marked as having 0 sessions, as have all students in the control group. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. Column (1) instruments sessions in ALEKS with the randomized allocation. Column (2) presents OLS value-added models in the full sample. Column (3) presents OLS value-added models using only data on the lottery-winners. One hour is equivalent to 60 minutes of platform use.

Table A1: Lee Bounds - Single Regression Approach

	Midline Math Test		Endline Math Test	
	Lower Bound (1)	Upper Bound (2)	Lower Bound (3)	Upper Bound (4)
CAL Treatment (β_1)	0.161 (0.142)	0.247* (0.140)	0.235 (0.173)	0.381** (0.168)
CAL x Tutoring (β_2)	-0.185 (0.157)	-0.028 (0.167)	-0.261 (0.171)	-0.239 (0.179)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.062 (0.140)	0.133 (0.137)	-0.004 (0.151)	0.120 (0.154)
Observations (before trimming)	1225		1218	
Observations (after trimming)	1206		1199	

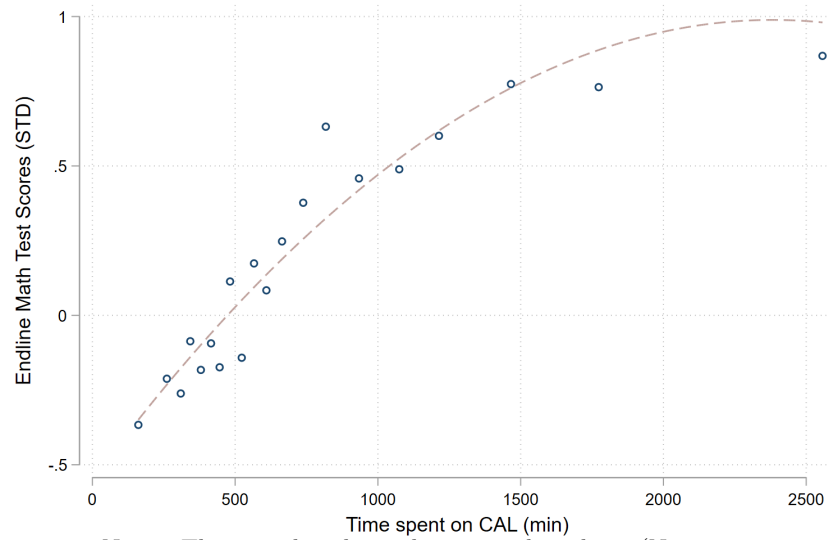
Notes: Lee (2009) bounds calculated using single regression approach with all three groups trimmed to minimum response rate (89.8%). Randomization inference standard errors in parentheses (1000 simulations). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ based on RI p-values.

Figure A1: Self-Efficacy Index by Performance Groups



Notes: The CAL + Tutoring group excludes tutees.

Figure A2: Returns to Hours of CAL Use



Notes: This sample is limited to treated students (N = 664).

A Additional Analysis

A.1 Power Calculations

To estimate statistical power, we simulate the randomization procedure described in Section 1.3 one thousand times and assume a minimum detectable effect (MDE) of 0.3 standard deviations. Evans and Yuan (2022) find this to be in the 80th percentile of effect sizes of RCTs with learning or access outcomes in low- and middle-income countries. Although this would be above the median of 0.2 for small RCTs, both tutoring and computer-adaptive learning have been identified as two of the most effective interventions to improve learning outcomes. A recent review of tutoring interventions by Nickow et al. (2020) found an overall pooled effect size estimate of 0.37 standard deviations. To the best of our knowledge, there are no meta-studies looking at effects from computer-*adaptive* learning technologies such as the one used in this study.

For this exercise, I use 2019 administrative data on 9th grade students in the same sample of schools selected for this intervention. This data includes students' test scores on a national diagnostic assessment, their sex, age, and a socioeconomic indicator. We also simulate and include a baseline test score variable. These simulations give us a statistical power of approximately 80 percent, which is the standard for randomized controlled trials in the social sciences. We aim to maximize this number by collecting prognostic variables at baseline, including tests scores on a short math assessment and other student characteristics. In Table A2, we present the results of the simulations using alternative MDEs (0.25, 0.3, 0.35).

MDE	Power
0.25	0.674
0.30	0.799
0.35	0.893

Table A2: Power Simulations

B Additional Results

This appendix presents detailed results for secondary outcomes summarized in Section 5.4.

B.1 Intrinsic Motivation, Absenteeism, and Dropout

Beyond academic performance, improvements in learning could also positively influence students' retention in school by increasing their motivation and interest in the subject, as well as their chances of obtaining a passing grade. In Tables B1, B2 and B3, we present impacts on intrinsic motivation, preferences towards math, absenteeism, and student dropout. Student dropout is a significant concern in our context, especially among the targeted age group. In the Dominican Republic, over 40 percent of students drop out between the 8th and 12th grades (IDEICE, 2017). In the control group, 8 percent of students dropped out of school during the intervention period (see Table B3). However, we find no evidence that the intervention impacted students' intrinsic motivation, how much they like math (self-reported 1 to 4 scale), or school absenteeism, or dropout rates by endline. The CAL+Tutoring group shows a marginally significant reduction in dropout rates ($p < 0.10$), though no effect is found for the CAL-only group. While this finding should be interpreted cautiously given it is only significant in one specification, it is consistent with the possibility that combining CAL and tutoring may reduce non-academic barriers to retention. The null effect on dropout is consistent with the possibility that, in this context, students face binding non-academic constraints—such as economic pressures or family obligations—that learning gains alone cannot address. The lack of findings on intrinsic motivation align with Araya et al. (2019), who also observed no effect on this outcome in a study using a game-based learning platform in Chile.

B.2 Math Self-Concept

As students spend more time on the software and master new topics, they receive new information on their abilities and may update their beliefs accordingly. In Table B4, we present effects on students' math self-concept, defined as students' perceptions of their own ability in mathematics compared to others. As in Araya et al. (2019), despite the large increases observed in math performance, we do not find statistically significant differences across experimental groups in a math self-concept index or on individual measures used to build the index. However, students in the CAL+Tutoring group were 9.3 percentage points less likely to agree with the statement “Math is harder for me than for many of my classmates” than students in the control group or in the CAL-only group. While these results are not conclusive, they may be suggestive of shifts in students' self-perception and confidence in their math abilities. Despite performing worse on math tests, the tutoring group seems to feel equally confident about their performance and are more likely to have mismatched expectations about their outcomes in the test, when compared to those whose

classrooms were not assigned to the tutoring component.

B.3 Self-Efficacy, Grit, and Perseverance

By offering personalized learning pathways and continuous feedback on individual progress and milestones, the platform allows students to understand and learn from their mistakes privately, without the stigma of public correction. As students master new topics and receive real-time feedback, this process may enhance their sense of competence and self-efficacy. In this subsection, we examine four related outcomes: self-efficacy, grit, perseverance, and locus of control. As shown in Table B5, neither the CAL intervention nor the tutoring intervention had a significant impact on a self-efficacy index.¹⁸ This suggests that the interventions did not notably alter students' beliefs in their ability to achieve academic success.

Similarly, we do not find impacts on a measure of grit (see Table B6), which assesses students' perseverance and passion for long-term goals (Duckworth et al., 2007). Following Alan et al. (2019), we implement an alternative measure of perseverance, based on a real-effort exercise. Table B7 shows that treated students are 9.38.7 percentage points more likely to choose to undertake an easier task against a more challenging alternative. We do not observe differences in their likelihood of giving up after failure for either group. This suggests that the adaptive nature of the platform, which presents exercises based on students' current ability, may have negatively impacted their willingness to take on more challenging exercises.

B.4 Student Wellbeing

To assess changes on students' wellbeing, we use two proxies: the Children's Depression Screener (ChilD-S) and a short questionnaire on math anxiety. For the first one, we build an index based on students' answers to 8 items about their mental state (e.g., "I feel happy", "I feel lonely", "It is hard for me to concentrate"). For the second one, we use two questions on students' worry about their math grades and their feelings before math tests to build a math anxiety index. Table B9 shows that the intervention had no impact on the first measure of wellbeing. While we do not find impacts on a math anxiety index, we find important differences when looking at individual measures of math anxiety. As seen in Table ??, students in the CAL-only group were 6.36 percentage points more likely to get nervous before math tests than students in the control and CAL+Tutoring groups. This difference is marginally significant ($p < 0.10$) statistically significant at the 5 percent level. While math anxiety is typically linked to lower performance (Dowker et al., 2016), this result may be positive if it signals higher levels of awareness or concern about their math performance in this group.

¹⁸This index is built using students' answers to the following items: 1) I can always solve difficult problems if I try hard enough, 2) I can solve most problems if I invest the necessary effort, 3) I can stay calm when I face difficulties because I can trust my coping skills, 4) When faced with a problem, I can usually find several solutions, and 5) If I'm in trouble I can find a solution.

B.5 Teacher Practices

The CAL platform may free up teachers' time by automating tasks such as developing and grading homework and tests. Additionally, the presence of tutors in CAL+Tutoring classrooms may affect teacher behavior through increased accountability or organizational demands. Table B11 presents effects on indices of teacher time mismanagement and classroom engagement, constructed from student reports.

We find that classrooms assigned to CAL+Tutoring showed marginally lower significantly lower teacher time mismanagement relative to the control group (0.23 SD without controls, $p < 0.10$; 0.18 SD with controls, not statistically significant 0.23 SD without controls, $p < 0.05$; 0.18 SD with controls, $p < 0.10$). The time mismanagement index captures whether teachers were late to class, absent from school, started or finished class early or late, or left school early. The significant reduction in this index suggests that teachers in CAL+Tutoring classrooms were more punctual and had fewer absences. This pattern may reflect increased accountability from the presence of tutors, or the greater organizational structure required to coordinate the combined intervention.

For teacher engagement, CAL-only classrooms showed marginally lower scores relative to control (0.18 SD, $p < 0.10$ without controls), though this effect is not robust to the inclusion of covariates. This pattern may reflect that the platform substitutes for some direct teacher instruction during CAL sessions. We do not observe significant differences in teacher engagement for the CAL+Tutoring group.

Platform data provide additional context: while teachers spent an average of 33 minutes per week actively using the platform, they seldom leveraged it to automate routine tasks. Over the course of the academic year, teachers assigned no more than three activities through the CAL platform, suggesting limited adoption of the platform's labor-saving features.

B.6 Student Time Use

One potential concern with technology-based interventions is that they might displace other productive activities or increase screen time outside of school. To examine whether the intervention affected how students allocate their time, we collected data on hours spent in eight categories: in-person classes, unpaid work at home, unpaid work outside the home, paid work, studying outside of school, meeting friends, entertainment, and sports.

Table B12 presents treatment effects on student time use. We find no statistically significant effects on any of these categories for either the CAL-only or CAL+Tutoring groups. The null effects suggest that the intervention operated primarily through the substitution of regular math instruction during school hours rather than through changes in how students allocated time outside of CAL sessions.

Table B1: Treatment Effects on Student Motivation and Attitudes

	Intrinsic Motivation (1)	Math Enjoyment (2)	Math Confidence (3)	Math Importance (4)
CAL Treatment (β_1)	-0.0218 (0.0829)	0.0363 (0.0683)	-0.0059 (0.0874)	-0.1416** (0.0626)
CAL $\times$ Tutoring (β_2)	0.1016 (0.0925)	0.0769 (0.0772)	-0.0137 (0.0962)	0.0943 (0.0848)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0798 (0.0802)	0.1132* (0.0672)	-0.0197 (0.0791)	-0.0473 (0.0600)
N	1194	1191	1179	1181
Control Mean	-0.003	2.731	2.331	2.746
Adjusted R ²	0.101	0.084	0.037	0.065

Notes: Randomization inference standard errors in parentheses (1000 simulations). All regressions include baseline covariates: student gender, age, baseline math score, parental employment status, and student confidence in math. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B2: Treatment Effects on Math Preferences

	Midline		Endline	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	-0.0430 (0.0868)	0.0109 (0.0732)	-0.0163 (0.0868)	0.0227 (0.0831)
CAL $\times$ Tutoring (β_2)	-0.0225 (0.1079)	-0.0330 (0.0847)	0.0243 (0.1078)	0.0241 (0.0970)
CAL + Tutoring ($\beta_1 + \beta_2$)	-0.0655 (0.0815)	-0.0221 (0.0696)	0.0079 (0.0810)	0.0468 (0.0774)
N	1218	1218	1210	1210
Control Mean	2.787	2.787	2.673	2.673
Adjusted R ²	-0.001	0.322	-0.002	0.235
Covariates	No	Yes	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B3: Treatment Effects on Absence and School Dropout

	Absence		Dropout	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	-1.4203 (2.1908)	-0.6303 (1.7159)	-0.0234 (0.0232)	-0.0142 (0.0186)
CAL $\times$ Tutoring (β_2)	1.9477 (2.4394)	1.1173 (1.8967)	-0.0059 (0.0285)	-0.0133 (0.0238)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.5275 (2.0593)	0.4870 (1.6236)	-0.0293 (0.0209)	-0.0274* (0.0165)
N	1333	1333	1333	1333
Control Mean	10.614	10.614	0.077	0.077
Adjusted R ²	0.002	0.138	0.001	0.056
Covariates	No	Yes	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Attendance measures are specific to math classes. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B4: Treatment Effects on Math Self-Concept

	Math Self-Concept Index	I usually do well in math	Math is harder for me than for many of my classmates
	(1)	(2)	(3)
CAL Treatment (β_1)	-0.0065 (0.0376)	0.0010 (0.0821)	0.0302 (0.0685)
CAL $\times$ Tutoring (β_2)	0.0127 (0.0408)	0.0032 (0.0935)	-0.1029 (0.0769)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0062 (0.0348)	0.0042 (0.0803)	-0.0727 (0.0624)
N	1184	1182	1175
Control Mean	0.693	2.658	2.523
Adjusted R ²	0.160	0.164	0.063

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B5: Treatment Effects on Self-Efficacy Index

	Midline		Endline	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	-0.0134 (0.0150)	-0.0155 (0.0124)	-0.0081 (0.0130)	-0.0114 (0.0115)
CAL $\times$ Tutoring (β_2)	0.0179 (0.0175)	0.0196 (0.0127)	-0.0011 (0.0174)	0.0045 (0.0157)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0046 (0.0134)	0.0041 (0.0117)	-0.0091 (0.0135)	-0.0069 (0.0133)
N	1217	1217	1194	1194
Control Mean	0.765	0.765	0.721	0.721
Adjusted R ²	0.000	0.248	-0.001	0.129
Covariates	No	Yes	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B6: Treatment Effects on Grit Index

	Endline	
	(1)	(2)
CAL Treatment (β_1)	0.0104 (0.0134)	0.0051 (0.0121)
CAL $\times$ Tutoring (β_2)	-0.0003 (0.0152)	-0.0004 (0.0135)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0101 (0.0127)	0.0047 (0.0117)
N	1333	1333
Control Mean	0.470	0.470
Adjusted R ²	-0.001	0.033
Covariates	No	Yes

Notes: The outcome is a standardized variable based on students' answers to the following items: 1) The schoolwork I like the most is the one that makes me think, 2) If I think I will lose in a game, I do not want to continue playing, 3) If I set a goal and see that it's harder than I thought I lose interest, and 4) I work hard on tasks. Answer options ranged from 1 (Doesn't seem like me at all) to 5 (Very much like me). Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B7: Treatment Effects on Logic Task

	Easy (1)	Difficult (2)	Give Up (3)
CAL Treatment (β_1)	0.0929** (0.0459)	-0.0966** (0.0410)	0.0037 (0.0281)
CAL $\times$ Tutoring (β_2)	-0.0416 (0.0551)	0.0250 (0.0543)	0.0167 (0.0340)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0512 (0.0469)	-0.0716* (0.0409)	0.0204 (0.0253)
N	1211	1211	1211
Control Mean	0.433	0.380	0.187
Adjusted R ²	0.068	0.072	-0.001

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B8: Treatment Effects on Locus of Control

	Endline	
	(1)	(2)
CAL Treatment (β_1)	0.0105 (0.0087)	0.0078 (0.0081)
CAL $\times$ Tutoring (β_2)	-0.0246* (0.0111)	-0.0216* (0.0110)
CAL + Tutoring ($\beta_1 + \beta_2$)	-0.0141 (0.0084)	-0.0138* (0.0081)
N	1196	1196
Control Mean	0.667	0.667
Adjusted R ²	0.003	0.073
Covariates	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B9: Treatment Effects on Student Depression

	Depression Index	
	(1)	(2)
CAL Treatment (β_1)	0.0116 (0.0112)	0.0090 (0.0091)
CAL $\times$ Tutoring (β_2)	-0.0046 (0.0109)	-0.0047 (0.0089)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0070 (0.0097)	0.0043 (0.0079)
N	1205	1205
Control Mean	0.612	0.612
Adjusted R ²	-0.000	0.082
Covariates	No	Yes

Notes: The outcome variable is a standardized index based on students' answers to the following items: 1) I am happy, 2) I worry a lot, 3) I feel sad, 4) I get upset quickly, 5) I often think I did something wrong, 6) It's hard for me to concentrate, 7) I feel lonely, and 8) I am not happy. Answers range from 1 (Strongly disagree) to 4 (Strongly agree). Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year, and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B10: Treatment Effects on Math Anxiety

	I'm worried of having bad grades in math (1)	I get nervous before math tests (2)	Math Anxiety Index (3)
CAL Treatment (β_1)	0.0333 (0.0326)	0.0632* (0.0356)	0.1404 (0.0911)
CAL $\times$ Tutoring (β_2)	-0.0116 (0.0356)	-0.0630 (0.0390)	-0.1202 (0.0962)
CAL + Tutoring ($\beta_1 + \beta_2$)	0.0218 (0.0294)	0.0003 (0.0358)	0.0202 (0.0847)
N	1199	1183	1182
Control Mean	0.773	0.713	0.000
Adjusted R ²	0.028	0.036	0.036

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * p<0.1, ** p<0.05, *** p<0.01 based on RI p-values.

Table B11: Treatment Effects on Teacher Behavior Indices

	Time Mismanagement		Teacher Engagement	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	-0.0752 (0.1205)	-0.0247 (0.1155)	-0.1782 (0.1082)	-0.1430 (0.0915)
CAL $\times$ Tutoring (β_2)	-0.1532 (0.1169)	-0.1546 (0.1100)	0.0415 (0.1248)	0.0372 (0.1141)
CAL + Tutoring ($\beta_1 + \beta_2$)	-0.2284** (0.1143)	-0.1793 (0.1135)	-0.1367 (0.0963)	-0.1058 (0.0776)
N	1148	1148	1151	1151
Adjusted R ²	0.007	0.117	0.005	0.116
Covariates	No	Yes	No	Yes

Notes: Randomization inference standard errors in parentheses (1000 simulations). Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ based on RI p-values.

Table B12: Treatment Effects on Student Time Use

	In person classes (1)	Unpaid work (at home) (2)	Unpaid work (out of home) (3)	Paid work (out of school) (4)	Study (5)	Meeting friends (6)	Entertain- ment (7)	Sports (8)
CAL Treatment (β_1)	0.0137 (0.2324)	-0.2492 (0.2158)	-0.0116 (0.2136)	0.0393 (0.2400)	0.0577 (0.1465)	0.0283 (0.1381)	0.0444 (0.0911)	0.0478 (0.0955)
CAL $\times$ Tutoring (β_2)	-0.2701 (0.2708)	-0.0167 (0.2254)	-0.1548 (0.2199)	-0.2637 (0.2486)	0.0460 (0.1495)	-0.1452 (0.2028)	-0.0180 (0.1025)	0.0013 (0.1166)
CAL + Tutoring ($\beta_1 + \beta_2$)	-0.2563 (0.2433)	-0.2660 (0.1878)	-0.1664 (0.2067)	-0.2244 (0.2187)	0.1037 (0.1365)	-0.1170 (0.1233)	0.0264 (0.0845)	0.0491 (0.0938)
Control Mean	6.01	3.80	2.27	2.73	2.03	2.24	3.33	2.08
Adjusted R ²	0.06	0.05	0.11	0.11	0.01	0.04	0.01	0.13
Observations	1182	1168	1156	1152	1177	1169	1169	1166

Notes: * $p < 0.01$, ** $p < 0.05$, *** $p < 0.1$. Covariates include student's gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. Randomization inference standard errors in parentheses (1000 simulations).